Frontier



Nordran GmbH (formerly Everest Carbon)

R&D application 2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Nordran GmbH (formerly Everest Carbon)

Organization location (we welcome applicants from anywhere in the world)

Blücherstraße 22, 10961 Berlin, Germany

Name(s) of primary point(s) of contact for this application

Pascal Michel

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Nordran GmbH is a Berlin-based for-profit company developing a novel alkalinity sensor. Formerly Everest Carbon GmbH, the company started in Enhanced Rock Weathering MRV and has since expanded into chemical monitoring applications for data centers.

1. Public summary of proposed project[1] to Frontier

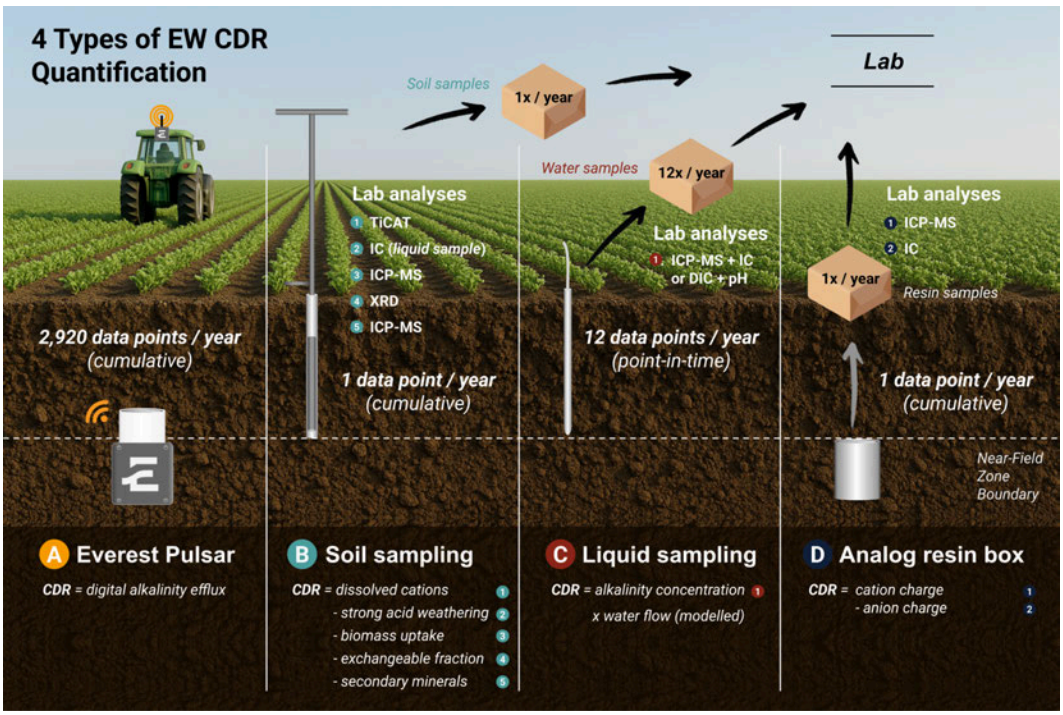
a. **Project title** <20 words

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

Field Calibration of In-Situ Alkalinity Sensors for Scalable ERW MRV

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Enhanced Rock Weathering (ERW) is a promising open-system carbon dioxide removal (CDR) pathway, with gigaton-scale long-term potential through deployment on existing agricultural land. However, its growth is constrained by a central Monitoring, Reporting, and Verification (MRV) bottleneck: the lack of affordable, field-deployable tools that can empirically quantify carbonate alkalinity export under real-world conditions. High-quality empirical field evidence of CDR at commercial scale is critical to further building credit buyer trust as well as model calibration datasets that will unlock a model-led future that allows ERW to deliver low-cost large-scale carbon removal. Today, most ERW projects rely on sparse porewater sampling, periodic laboratory analysis, and hydrological models to derive aqueous phase CDR estimates. These approaches are labor-intensive, expensive, and often fail to capture the strong temporal variability of alkalinity export driven by variance in rainfall, infiltration, and soil hydrology. As a result, uncertainty remains high and monitoring costs make ERW less attractive at commercial scale.

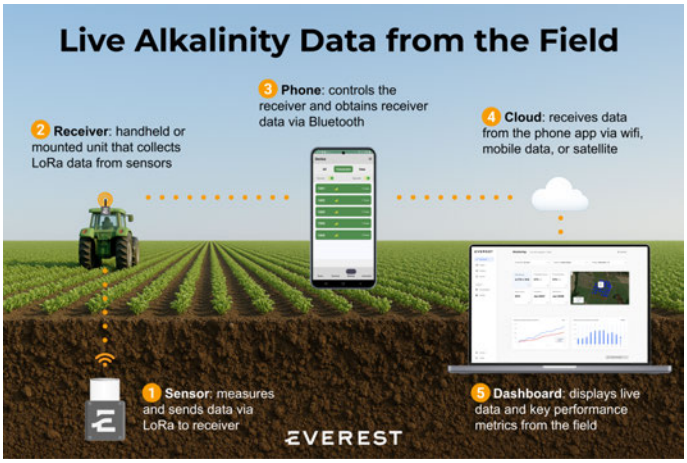


Our project targets Frontier's priority innovation area, "**Improving the MRV toolkit for open-system pathways**," by field-calibrating the first scalable system for direct, continuous measurement of

cumulative alkalinity export from ERW soils. This work advances the system to crediting readiness for deployment in commercial ERW projects.

Our objective is to equip ERW researchers and project developers with a sensor that quantifies alkalinity generation and transport through soils with continuous temporal resolution. This enables, for the first time, robust model-based MRV at scales relevant to meaningful climate impact.

To achieve this, we have developed a novel in-situ sensor, Pulsar, which uses a weak-acid ion exchange resin to selectively capture alkalinity species from percolating soil water. A proprietary transduction mechanism converts this cumulative uptake into a stable digital signal, providing a direct and time-integrated measure of alkalinity export. In controlled lab soil-column experiments, the system demonstrated **98.9 ± 0.3% alkalinity capture efficiency without soil, 97.7 ± 0.2% with soil, and digital readout agreement within -0.3 ± 6.0% relative deviation**, establishing proof of the core sensing mechanism ["Direct In Situ Measurement of Alkalinity Export for Real-Time Enhanced Weathering MRV", Muth et al. (2025)].



The remaining barrier to commercial MRV deployment is not sensing chemistry or physics, but **hydrological calibration**: how measured alkalinity passing through the device translates into a total field-scale alkalinity export measurement under heterogeneous agricultural conditions. Soil texture, compaction, moisture dynamics, rainfall intensity, and installation geometry determine what fraction of field percolation is intercepted by the sensor. Without precise calibration of this relationship, uncertainty in field-scale carbon removal claims remains too high for broad crediting adoption.

The core goal of this proposed Frontier project is therefore to resolve this bottleneck through a rigorous **field-scale calibration campaign of our alkalinity sensors**.

We will do this through three integrated workstreams:

1. Sensor interception optimization

We will refine the device architecture to ensure a consistently high and well-characterized capture fraction of local percolation across representative agricultural soils, preventing hydrological model uncertainties from compounding to unacceptable levels.

2. Tracer-based hydrological calibration

Using conservative bromide pulse tracer experiments and time-staggered sensor extraction campaigns, we will directly quantify effective water interception fractions under field conditions.

3. Hydrological modeling and validation

We will use empirical tracer data to parameterize and validate a HYDRUS-3D model of Pulsar performance across diverse soil textures and ERW microplot deployments. This will enable the development of a transferable calibration function that maps sensor output to true hectare-scale alkalinity export, conditioned on site-specific soil properties and climate.

By project completion, we aim to demonstrate:

- Hydrologically-calibrated sensors measuring field alkalinity export with **<10% hydrology-induced uncertainty** as evidenced by bromide tracer datasets
- Robust sensor performance across multiple representative soil classes
- Continuous datasets across three different soil textures revealing temporal alkalinity export dynamics unavailable through discrete sampling
- A deployable sensor MRV framework for large-scale ERW projects
- A manuscript with open datasets of our validation field trials in peer-review for publication

If successful, this project would unlock a **step-change in cost and scale for ERW MRV**.

Instead of relying on manual liquid sampling and expensive lab workflows, developers could deploy dense networks of autonomous buried sensors that continuously generate cumulative alkalinity data. This **helps lower monitoring cost significantly today**, and most importantly, **unlocks the required high temporal resolution** alkalinity generation and migration datasets required **for model-led MRV systems**.

The broader **CDR relevance extends beyond ERW**. Many open-system pathways, including ocean alkalinity enhancement and inland water alkalinity enhancement, face the same challenge of measuring dissolved alkalinity fluxes in dynamic environments. A successful outcome here would establish a new class of field-deployable MRV infrastructure applicable across multiple carbon removal pathways.

In summary, **Frontier funding would help transform alkalinity monitoring from an expensive niche workflow into scalable infrastructure** for open-system CDR verification. That capability is essential if ERW and related pathways are to move from pilot projects to model-centric climate-relevant deployment.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

This project will establish the first field-calibrated framework for using autonomous in-situ alkalinity sensors as a quantitative MRV tool for Enhanced Rock Weathering (ERW). Prior work validated the sensing mechanism in controlled environments, demonstrating selective alkalinity capture, high efficiency, and accurate digital readout. **The remaining challenge is hydrological calibration**: translating sensor measurements into robust hectare-scale alkalinity export under heterogeneous field conditions. Soil texture, compaction, preferential flow, moisture regime, rainfall intensity, and installation geometry all influence the fraction of percolation captured. This project will quantify, predict, and reduce that uncertainty to levels compatible with commercial MRV methodologies. **Resolving these uncertainties enables credit-grade, continuous alkalinity monitoring and model-centric ERW MRV at scale.**

Key research questions

We will answer four core questions:

1. How can sensor design be improved to maximize representative interception of field percolation?

Current prototypes may intercept only a fraction of local percolation under some conditions. We will test inlet geometries, permeable interfaces, and installation methods to increase interception toward unity and reduce variance across soils.

2. What fraction of surrounding field water flux is actually sampled by the sensor under real conditions?

We will use bromide tracer pulse experiments with time-staggered sensor retrievals to quantify interception across soils, installation methods, moisture states, and infiltration events.

3. Can a mechanistic hydrological model be reliably calibrated with empirical sensor interception measurements to generalize across sites and climates?

Tracer datasets will refine HYDRUS-3D models linking rainfall, soil texture, and installation to measured interception. We will test agreement between predictions and observations to quantify remaining uncertainty.

4. Can the combined system reduce MRV uncertainty to levels suitable for commercial ERW crediting?

Refined sensors and calibrated models will be deployed in ERW treatment and control microplots across multiple soils to assess whether hydrology-induced uncertainty meets crediting requirements.

Proposed approach

The project is structured into four integrated workstreams, progressing from hardware optimization to

empirical calibration, predictive modeling, and ERW field validation. The **18-month program** will be coordinated from Berlin, with deployments **across partner sites in Germany, Austria, Spain, and Slovenia** covering diverse ERW-relevant soils and climates.

Workstream A: Sensor hydrological optimization and indicative field pre-screening

We will first optimize the sensor's hydraulic interface and matrix through iterative soil-column and small-plot infiltration tests. Indicative bromide tracer recovery tests at our partner site near Berlin will serve as a design-screening tool to confirm that candidate architectures achieve sufficient interception efficiency to advance into field calibration.

Mechanical refinements will target key drivers of interception, including higher-permeability cartridge layers, optimized inlet and outlet membrane area, reduced internal hydraulic resistance, improved flow-distribution geometry, anti-clogging interfaces, minimized bypass flow, elimination of air gaps, and installation methods that reduce smearing, compaction, and preferential flow paths.

Indicative bromide tracer pulse field tests will be used to down-select and validate the most promising architectures. Two installation methods will be evaluated: vertical insertion using an electric auger, and horizontal installation via trenching beneath an undisturbed soil column. The primary output is a refined device design ready for multi-site field calibration.

Validation will be conducted using 100 analog prototype devices spanning two installation methods and five architecture variants, deployed in sandy loam soils representative of common ERW settings. Devices will undergo repeated infiltration cycles under both controlled irrigation and natural rainfall to capture performance across realistic hydrological regimes.

Workstream B: Two-phase empirical tracer calibration

We will conduct bromide tracer pulse experiments to quantify the fraction of field percolation captured by the sensor under realistic conditions, how this changes after installation, and how transferable the calibration is across soils and sites. This workstream is integrated with Workstream C, with tracer results informing hydrological model refinements and model outputs guiding subsequent field trials.

Phase 1: Post-installation equilibration and baseline capture fraction

The selected sensor geometry will first be deployed across **two representative soil textures** (sandy loam and sandy clay loam), with all devices buried simultaneously. Bromide tracer pulses will then be applied at staggered intervals after deployment, with matched time-staggered sensor retrievals over approximately **4-8 weeks**, depending on natural precipitation.

This design allows us to get an initial measure of interception fraction as a function of time since installation. Burying sensors temporarily disturbs local soil structure and flow paths. Even though re-equilibration will likely not yet be complete after 8 weeks, this first campaign will give us a

quantitative indication of its rate, as well as an extrapolated indication of its terminal value.

Interim refinement window

Following Phase 1, we will complete a short optimization cycle using the first field results. This will include targeted mechanical refinements to the sensor geometry, adjustments to installation procedures, and updates to deployment SOPs designed to further improve hydrology and reduce operator variability before the second campaign.

Scale: 80 analog prototype devices across 2 sites, 4 extraction timepoints and 2 installation methods.

Phase 2: Generalization and longer-duration validation

A second tracer campaign will be conducted across **three sites and soil textures**, including two previously studied soils and one new site and soil. This phase will run for approximately **3 to 5 months**, capturing a broader range of rainfall, infiltration, and seasonal moisture conditions.

In parallel with Workstream C, Phase 2 will assess how well the calibrated framework generalizes across repeated deployments in known soils and varying conditions, as well as to a previously unseen soil without prior calibration.

Tracer results will be continuously incorporated into the HYDRUS-3D model, with predictions compared against observed recovery. This iterative process will enable final refinement of transferable correction functions before integrated ERW validation.

Scale: 192 analog prototype devices across 3 sites, 8 extraction timepoints and 2 installation methods.

Workstream C: Hydrological modeling

In parallel with Workstream B, we will iteratively calibrate and refine existing **HYDRUS-3D** models that simulate water flow around the buried sensor. The objective is to use empirical tracer data to build a transferable hydrological model that translates sensor measurements into field-scale alkalinity export with low uncertainty.

Initial models capture expected flow behavior around the current sensor geometry. Phase 1 tracer data will calibrate key parameters, including near-device hydraulic conductivity, soil–sensor contact resistance, preferential flow and bypass, and time-dependent equilibration after installation.

After Phase 1, model outputs will guide refinement of mechanical design and installation SOPs. In Phase 2, predictions will be validated against longer-duration tracer datasets across both known and unseen soils to assess generalization beyond initial calibration conditions.

Following Phase 2, a final refinement will produce the model used in Workstream D. We target agreement between modeled and measured tracer recovery `within ≤20%`, with a `stretch target of ≤10%`.

Workstream D: ERW microplot validation and integrated MRV readiness

Finally, the refined sensor design, deployment SOP, and calibrated HYDRUS-3D framework will be validated in ERW microplot trials over a minimum six-month field season.

Replicated treatment and control microplots will be established across representative soils. Treatment plots will receive finely milled basalt under standard agronomic practice, while controls remain otherwise identical. Sensors will remain buried throughout the trial to continuously measure cumulative alkalinity export.

Reference measurements will include porewater leachate chemistry, rainfall, and soil moisture monitoring.

By month 18, this workstream will deliver the first integrated field validation of a real-time alkalinity sensor for ERW MRV, targeting hydrology-induced uncertainty below 10%. Results will be consolidated and published, including open-access datasets covering field-scale bromide tracer experiments, alkalinity measurements, and reference leachate analyses, in a peer-reviewed journal.

Scale: 84 Pulsar sensors, 36 suction cup lysimeters and 24 soil moisture probes across three sites.

Why our approach is best-in-class

Our approach is best-in-class because it directly measures the cumulative quantity of interest, aqueous carbonate alkalinity export, continuously and in situ, rather than inferring it from proxies such as mineral weathering. Established methods rely on intermittent sampling, which undersamples systems governed by highly non-linear hydrological dynamics.

Resin-based approaches typically lack temporal resolution, as measurements require excavation and lab analysis. Pulsar's novel readout enables continuous data capture, providing increased visibility into alkalinity export dynamics. This higher-resolution data supports models that link environmental drivers to alkalinity transport rates.

Pulsar also offers a cost trajectory that alternatives cannot match. Hardware costs can decrease significantly with scale, while field operations do not benefit from comparable efficiencies. This makes Pulsar the most viable path toward $100/tCO2 ERW credits based on representative in-situ measurements.

Differentiation from other organizations

Nordran's leadership team brings a distinct mix of expertise compared to other ERW players, spanning physics, electrical engineering, sensor design, machine learning, and industrial engineering. This combination provides the technical depth required to develop novel MRV solutions.

As a pure technology company, Nordran can deploy measurement solutions more rapidly. MRV innovations developed by project operators are often kept proprietary to preserve cost or credit advantages, limiting industry adoption. Nordran, by contrast, is incentivized to scale its solutions broadly.

Nordran's prior experience developing ERW projects in India is an additional differentiator. The team has direct exposure to field constraints and understands the operational realities faced by customers.

This experience, combined with a long track record in the space, supports credibility and adoption, showcased by existing pilots with more than 10 project developers on five continents.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Scientific risk - hydrological calibration and signal attribution.

Uncertainty in how well the sensor captures field-scale percolation, and how clearly ERW-induced alkalinity can be separated from background variability in alkalinity export, could limit quantification accuracy.

Mitigation: Multi-site trials across different soil textures, Hydrus-3D modelling of the actual hardware architecture and iterative empirical validation using conservative bromide tracer studies.

Project execution risk - multi-site coordination and data completeness.

Delays in site readiness, installation, or inconsistent data collection across partners could reduce statistical power.

Mitigation: Modular deployment design (reallocating sensors across sites if needed), standardized protocols, centralized data pipelines, and clear milestone-based go/no-go gates.

MRV risk - insufficient accuracy or acceptance for crediting.

Sensor outputs may not meet required accuracy thresholds or be accepted by registries and buyers for quantification.

Mitigation: Benchmarking against reference methods (suction cup lysimeters + lab analyses), conservative uncertainty treatment, early engagement with registries, and phased positioning from secondary validation to primary quantification.

Ecosystem risk - slow market and standards adoption.

Even with strong results, ERW MRV standards and buyer demand may evolve slowly, delaying commercial uptake.

Mitigation: Active participation in methodology discussions, publication of results, alignment with leading registries, early pilot deployments with project developers as well as market diversification.

Financial risk - insufficient resources for full-scale validation.

Field trials and hardware iteration are capital-intensive, and lack of funding could constrain scope or

speed.

Mitigation: Focused project design with clear milestones, leveraging existing infrastructure and partnerships, and using Frontier funding to enable dedicated execution as well as sustainable follow-on ramp-up of commercial volumes.

Other risks - regulatory and permitting delays.

Approvals for tracer studies or site activities could delay timelines.

Mitigation: Early engagement with authorities, use of established protocols, and flexibility in site selection and sequencing.

Overall, risks are actively managed through a combination of experimental design, iterative engineering, and staged validation.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	**Optimized field-ready Pulsar sensor design** with crediting-grade hydrological calibration, validated installation SOPs, and demonstrated interception performance across representative soils. **Two-phase bromide tracer calibration dataset** quantifying sensor interception fraction, post-installation soil equilibration dynamics, and transferability across multiple sites and soil textures. **Field-validated HYDRUS-3D model** translating sensor measurements into field-scale alkalinity export measurements with quantified, low uncertainty bounds for new deployments. **ERW microplot validation dataset** demonstrating continuous alkalinity export measurement under finely milled basalt application and benchmarking against conventional reference methods. **Peer-reviewed manuscript** describing the first field validation of an in situ alkalinity sensor for ERW MRV.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader*	This project resolves the central MRV bottleneck in Enhanced Rock Weathering: the lack of scalable,

CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)	empirical measurement of alkalinity export under real field conditions. By replacing sparse, labor-intensive sampling with continuous, time-integrated in-situ measurements, it enables **a step-change in both measurement fidelity and data density.** Crucially, this unlocks a transition to model-led MRV, which is essential for scaling ERW to climate-relevant levels. Direct measurement alone cannot scale to millions of hectares due to cost and operational constraints. Instead, scalable MRV requires predictive models that estimate carbon removal across large areas, continuously calibrated and constrained by real field data. **This project provides the missing data layer for the transition to model-led MRV**. High-frequency, spatially distributed sensor datasets enable models to be trained, validated, and audited with unprecedented resolution, reducing uncertainty while avoiding dense and costly manual sampling. Together, this shifts ERW MRV from episodic measurement to scalable monitoring infrastructure, enabling low-cost verification and large-scale deployment.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	**Month 1-4: Optimized sensor design selected for field calibration** Mechanical refinements, controlled infiltration testing, and indicative tracer screening at the Berlin-area partner farm. Final candidate geometry selected for formal calibration. **Month 4-8: Initial field interception fraction quantified and first update to hydrological model completed** Two-site bromide tracer campaign measuring interception fraction and post-installation settling dynamics. Initial HYDRUS-3D calibration completed, together with targeted updates to sensor geometry and deployment SOPs based on first field results. **Month 8-13: Transferable multi-site calibration framework validated**

	Three-site longer-duration tracer campaign, including one previously unseen soil/site. Iterative model refinement completed and final transferable calibration framework produced. **Month 13-18: Integrated ERW field validation completed** Six-month treatment/control field trial using finely milled basalt across representative soils, validating the full MRV chain from buried sensor signal to field-scale alkalinity export estimate. **Month 16-18: Manuscript submitted for peer-review** Complete integrated validation dataset, finalized calibration methodology, and submission of peer-reviewed manuscript including open-access datasets. The project is complete when Pulsar has demonstrated a validated framework linking buried sensor measurements to field-scale alkalinity export with a target of **<10% hydrology-induced uncertainty**, supported by multi-site field data and publication-ready evidence.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	**$571,252**